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Remagnetization of the Coast Range Ophiolite and Lower Part of the Great Valley Sequence in Northern California and Southwest Oregon

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Overprinted magnetizations have been found at four localities in the Middle Jurassic Coast Range ophiolite and the overlying Upper Jurassic and Lower Cretaceous Great Valley sequence in northern California and at one locality in the partially correlative Lower Cretaceous Days Creek Formation in southwest Oregon. At Del Puerto Canyon, on the east side of the Diablo Range, a pilot study of the upper Jurassic Lotta Creek Formation gives in situ paleomagnetic directions grouped around the present geomagnetic field, suggesting magnetic overprinting long after deposition and folding. We suggest that the loss of original magnetization could be the result of long burial at about 7 km depth, followed by Late Tertiary uplift; the possibility of chemical remagnetization, however, can not be excluded. Paleomagnetic data from sandstones from the Lower Cretaceous Great Valley sequence in the Wilbur Springs area suggest Cenozoic remagnetization that could be related to Pliocene and Pleistocene volcanic and hydrothermal activity in the area. The results from calcareous concretions in the Wilbur Springs area, from 21 sites from basalts and overlying sedimentary rocks of the Great Valley sequence at Stonyford, and from several sites in various rock types in the Coast Range ophiolite and Great Valley sequence near Paskenta suggest unblocking of magnetization during the long period of burial at depth of 7 km or more and remagnetization during tectonic uplift in the Late Cretaceous or Tertiary. Chemical remagnetization, however, can not be excluded. The paleomagnetic data from six sites in Lower Cretaceous sedimentary rocks of the Days Creek Formation in southwest Oregon, which overlies a dismembered ophiolite, fail the fold test. In situ paleomagnetic directions group near the expected Tertiary field directions. These sedimentary rocks may have been remagnetized during a major Eocene tectonic event known to have occurred in this region. The paleomagnetic results confirm that the Coast Range ophiolite and the seemingly little-deformed Great Valley sequence in northern California, as well as correlative rocks in southwest Oregon have had a complex tectonic history. Once the nature and timing of these events are better understood, the timing of remagnetization may be further constrained, which, in turn, could give further insight into the nature of the tectonic events.

INTRODUCTION

The Great Valley of California is an elongate basin filled with Jurassic to Holocene sedimentary deposits (Figure 1). Along the west side of the valley, in the Coast Ranges a thick sequence of Upper Jurassic and Cretaceous sedimentary rocks, the Great Valley sequence, locally lies on a Middle Jurassic ophiolite sequence, the Coast Range ophiolite [*Bailey et al.*, 1970; *Hopson et al.*, 1981]. On the east side of the Great Valley, Upper Cretaceous strata of the Great Valley sequence lie unconformably on Jurassic and older basement rocks of the Sierra Nevada province [*Ingersoll*, 1982]. The nature of the concealed boundary between the Sierran basement on the east and the Coast Range ophiolite to the west is controversial and has been the subject of much recent geophysical research. *Cady* [1975] concluded that the Great Valley is underlain by a fragment of oceanic crust and suggested a model involving westward stepping of a Late Jurassic, eastward dipping subduction zone, with imbrication of the oceanic slab. More recently, it has been proposed [*Suppe*, 1978, 1979; *Griscom*, 1983; *Wentworth et al.*, 1984; *Ricci et al.*, 1985] that the Coast Range ophiolite formed many tens of kilometers west of its present position and was subsequently thrust (obducted) eastward onto Sierran-Klamath basement. West of the Great Valley, the Coast Range ophiolite and Great Valley sequence are faulted against the Franciscan assemblage of the Coast Ranges along the Coast Range thrust and successor faults [*Bailey et al.*, 1970; *Hopson et al.*, 1981].

The Great Valley sequence and underlying Coast Range ophiolite are tilted to the east along the western side of the Great Valley; the timing of this tilting is now well constrained. In the San Joaquin Valley, comparison of the tilt of Cretaceous and Tertiary sedimentary rocks suggests some deformation prior to and during Tertiary sedimentation [*Page*, 1981]. The unconformity between the Eocene Capay Formation and the underlying Cretaceous Great Valley sequence, southeast of Wilbur Springs, suggests some deformation prior to the Tertiary sedimentation in the Sacramento Valley too. A possible unconformity between the Lower and Upper Cretaceous Great Valley sedimentary rocks near Wilbur Springs [*Lawton*, 1956; *Rich*, 1971] suggests that tectonism occurred in the Middle Cretaceous; however, the apparent unconformity could also be the result of bedding-plane thrusts (D. L. Jones, oral communication, 1984). *Zieglar and Spotts* [1978] suggested that onlap relations between the Upper Cretaceous and Eocene sedimentary rocks are a result of Late Cretaceous and Early Tertiary uplift of the Coast Ranges. This is supported by the presence of undoubted Franciscan detritus in Early Tertiary (Paleocene?) conglomerate of the

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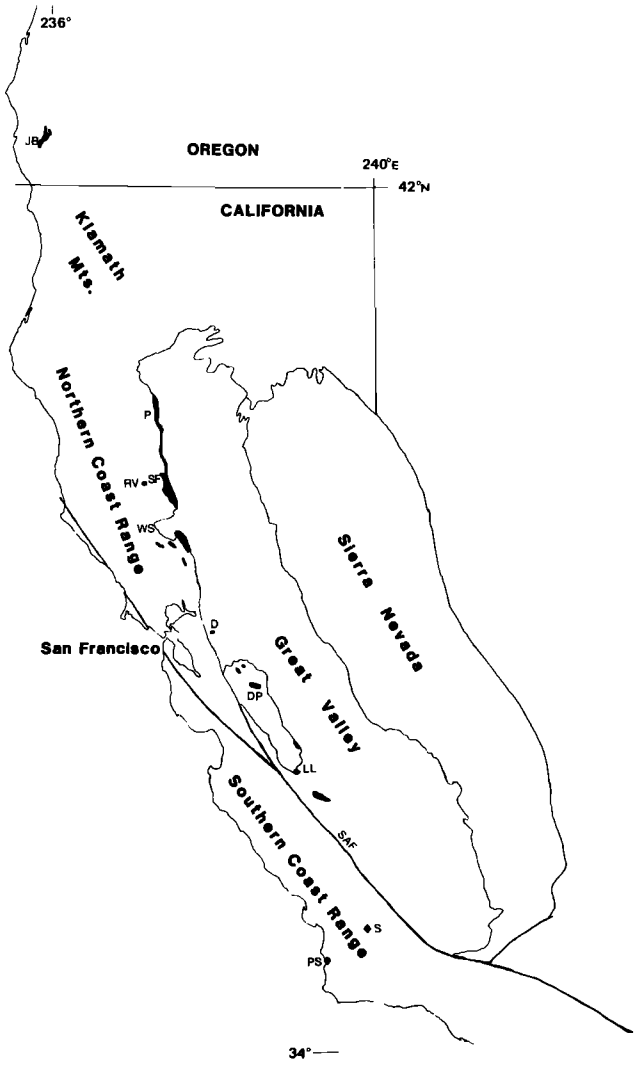


Fig. 1. Location of outcrops of the Coast Range ophiolite (colored in black). Localities of the Coast Range ophiolite and Great Valley sequence discussed in the text are lettered as follows: D, Mt. Diablo; DP, Del Puerto; JB, Jacoby Butte; LL, Llanada; P, Paskenta; PS, Point Sal; RV, Rice Valley; S, Stanley Mountain; SF, Stonyford; WS, Wilbur Springs; SAF, San Andreas fault.

Rice Valley outlier, 25 km west of Stonyford [*Berkland*, 1973; *Dickinson et al.*, 1979]. The major pulse of uplift of the northern Coast Ranges, however, appears to postdate about 3.4 Ma, as abundant Franciscan detritus appears in the late Cenozoic deposits of about this age in the northern Sacramento Valley [*Blake et al.*, 1984].

In southwest Oregon, there is no Great Valley as a geomorphic feature. Here, rocks correlative with the Coast Range ophiolite and Great Valley sequence are called the Snow Camp terrane and are thrust onto rocks of the Klamath Mountains province (correlative with the Sierra Nevada province to the south). Rocks of the Klamath Mountains are, in turn, thrust over the Dothan Formation, a partial correlative of the Franciscan assemblage [*Blake et al.*, 1985*a*].

Different tectonic models have been proposed for the environment of formation of the Coast Range ophiolite, ranging from open-ocean, deep-water, slow-spreading ridge [*Hopson et al.*, 1981] to intraarc rifting [*Schweickert and Cowan*, 1975; *Saleeby*, 1981; *Sharp and Evarts*, 1982; *Shervais and Kimbrough*, 1985]. Pb-U radiometric ages determined on plagiogranite from the Coast Range ophiolite in California fall mainly in the limited range of 169–161 Ma [*Hopson et al.*, 1981; *McLaughlin and Ohlin*, 1984]. Some slightly younger ages have been determined locally on dikes and volcanic rocks and may represent younger arc-related activity [*Blake and Jones*, 1981; R. Evarts, oral communication, 1984] that predates deposition of the sedimentary rocks of the Great Valley. In southwest Oregon, plagiogranite from ophiolite in the Snow Camp terrane gives a Pb-U age of 169 ± 1 Ma [*Saleeby et al.*, 1984].

Paleomagnetic studies by *McWilliams and Howell* [1982] of basalts and overlying sedimentary rocks near Stanley Mountain (Figure 1) and by *Luyendyk and Hornafius* [1982] on basalts at Point Sal (Figure 1) suggest that these ophiolitic fragments may have formed at lower latitudes than expected from their present position with respect to North America. Both these ophiolites are in the southern Coast Ranges and west of the San Andreas fault, but are traditionally correlated with the Coast Range ophiolite [*Bailey et al.*, 1970; *Hopson et al.*, 1981]. In a study of a sheeted complex at Mt. Diablo *Williams* [1983], concluded that there had been little or no latitudinal shift, but that the rocks had undergone a counterclockwise rotation. *Saad* [1969] studied ultramafic units in the ophiolite fragment at Red Mountain near Del Puerto in the northern Diablo Range, which had somewhat scattered directions of magnetization, possibly because the samples were partially serpentinized. Preliminary results based on 10 basalt cores from a small part of the section in the Llanada remnant of the Coast Range ophiolite [*Beebe and Luyendyk*, 1983] suggest a low-latitude origin for these rocks.

Paleomagnetic results from outcrops of the Upper Jurassic to Upper Cretaceous sedimentary rocks of the Great Valley sequence in the northern Sacramento Valley [*Mankinen*, 1978; *Mankinen and Irwin*, 1982] suggest remagnetization during the Late Cretaceous. *Ward et al.* [1983] report finding reversed chron 33r, an interval of reversed polarity from 83–79 Ma [*Harland et al.*, 1982], in Upper Cretaceous rocks of the Great Valley sequence on both sides of the Sacramento Valley. These results suggest that remagnetization in the northern part of the Great Valley, if widespread, must have either taken place before 83 Ma or else affected only strata older (and more deeply buried) than Campanian. In this study, we investigate the horizontal and vertical extent of the remagnetization in several well-dated localities in northern California and southwest Oregon and its possible causes.

Geologic Setting and Paleomagnetic Results

Oriented samples collected at five widely spaced localities from the Coast Range ophiolite and Great Valley sequence were mostly drilled in the field and oriented using a sun compass or back azimuths; at a few sites, oriented hand samples were collected. A short summary of the geologic setting, laboratory techniques, and results of the five sample localities follows.

Del Puerto Canyon

The Coast Range ophiolite in the vicinity of Del Puerto canyon (Figures 2*a* and 2*b*) consists of a basal harzburgite and dunite unit that structurally overlies the Franciscan assemblage along the Coast Range thrust [*Evarts*, 1977; *Blake*, 1981]. The harzburgite unit is overlain by locally feldspathic cumulate ultramafic rocks, which, along with the underlying harzburgite and dunite, are faulted against a slab of gabbro

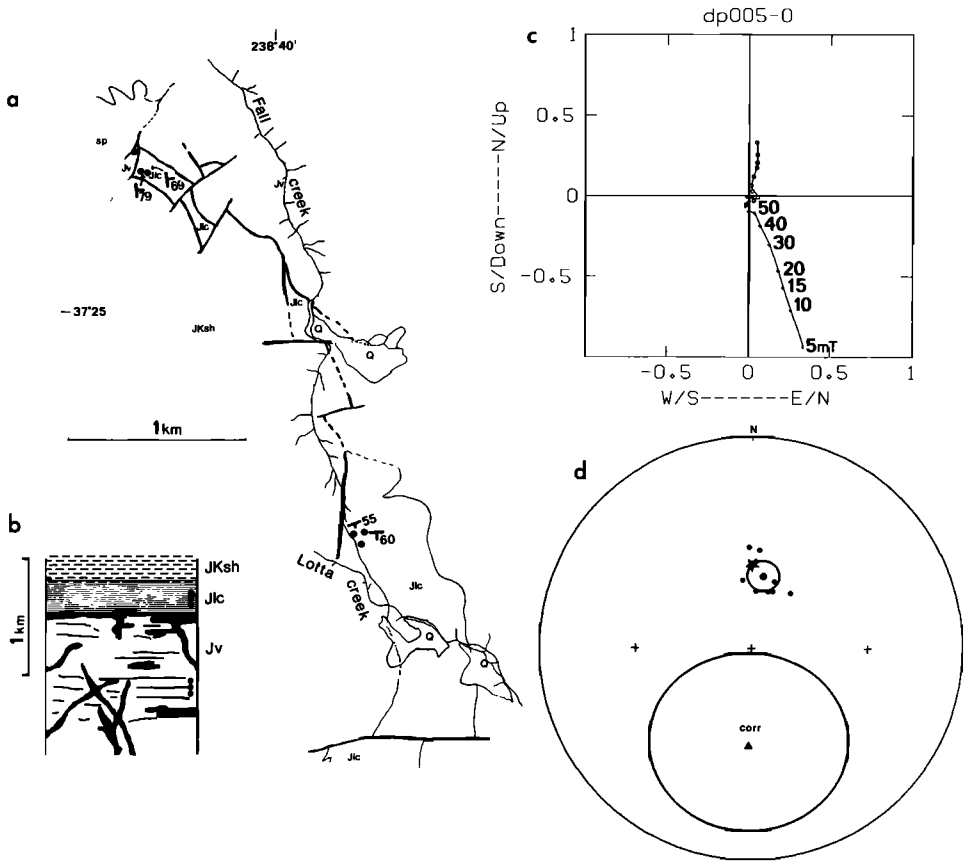


Fig. 2. Geologic setting and paleomagnetc results from Del Puerto Canyon. (*a*) Generalized geologic map modified from *Evarts* [1977]: Q, Quaternary; JKsh, shale unit of Great Valley sequence; Jlc, Lotta Creek Formation (volcaniclastic sandstones and tuffaceous cherts) of the Great Valley sequence; Jv, volcanic rocks unit of the Del Puerto ophiolite. Thick lines are faults; circles are sampling localities. (*b*) Part of the stratigraphic column sampled. (*c*) Projection of the measured magnetization during alternating magnetic fields treatment of a sample of volcaniclastic sandstone of Lotta Creek Formation. Circles are in horizontal plane, plus signs are in vertical plane. Axial length is 9.8×10^{-8} A m^2. (*d*) Paleomagnetic results on an equal area projection. Plus signs are at 45° inclination. Solid circles, clean magnetic directions of samples before structural correction; larger solid circle, mean, with oval of 95% confidence; solid triangle, mean of cleaned direction after structural correction, with large oval of 95% confidence; star, axial geocentric dipole field at Del Puerto.

intruded by rare minor plagiogranite dikes. This gabbro is structurally overlain by a volcanic sequence that is characterized by abundant quartz keratophyre [*Evarts*, 1977]. Hornblende from a feldspathic peridotite and from a hornblende gabbro yield K-Ar ages of 164 ± 8 and 162 ± 5 Ma, respectively [*Lanphere*, 1971, and oral communication, 1985]. Zircon from plagiogranite cutting the gabbro gives a Pb-U age of about 155 Ma [*Hopson et al.*, 1981]. Recent Ar-Ar dating of quartz-keratophyre from the volcanic sequence (W. Sharp, oral communication, 1984) also yields ages younger than the gabbro and, like the plagiogranite, probably represents island-arc volcanism that evolved above the ophiolitic basement. Tuffaceous chert and volcaniclastic sandstone of the Lotta Creek Formation overlie the volcanic sequence and are at the base of the Upper Jurassic (Tithonian) shale and sandstone of the Great Valley sequence. All of the ophiolitic, siliceous volcanic, and clastic rocks have been collectively assigned to the Del Puerto terrane [*Blake et al.*, 1982].

The Del Puerto terrane has undergone at least two important phases of faulting. Early imbricate thrust faults have been mapped between the Coast Range ophiolite and the Franciscan assemblage and within the ophiolite. Based on regional geologic data, they probably experienced movement during the Late Cretaceous (≈90 Ma; R. C. Evarts, oral communication, 1985). These thrust faults are cut by high-angle faults including the Tesla-Ortigalita fault of probably Neogene age [*Raymond*, 1973; *Page*, 1981].

Exposures in the Del Puerto area are poor and the volcanic rocks are weathered. As a pilot study, nine samples (1–3 specimens each) were collected from the sedimentary rocks of the Upper Jurassic Lotta Creek Formation in a west-dipping fault block near Fall Creek (Figure 2*a*). Four pairs of specimens (each pair from one sample) were cleaned, one specimen by stepwise thermal and the other by stepwise alternating magnetic field demagnetization; both procedures gave the same results. The remaining five samples were cleaned using alternating fields. Vector component diagrams show linear demagnetization trends, indicating the removal of a single component of magnetization (Figure 2*c*). A program modified after *Kirschvink* [1980] was used to fit a line to the demagnetization results. Thermal demagnetization experiments suggest that the magnetic carrier is magnetite.

The in situ results from these samples are all normal and very close to the axial geocentric dipole field at this locality. Restoring the magnetic vectors to the paleohorizontal of the sedimentary rocks increases the scatter (Figure 2*d*, Table 1), indicating that the sedimentary rocks were remagnetized after folding.

TABLE 1. Mean Field Directions

Site	In Situ Inc.	In Situ Dec.	Corrected Inc.	Corrected Dec.	N	κ	α_{95}	Strike	Dip
Del Puerto									
all samples	62.0	9.1			9	78	5.9	see map	
all samples			52.0	181.1	9	3	36.2		
Wilbur Springs sandstone									
cl001	71.0	34.0	55.9	223.5	5	127	6.8	130.0	53.0
cl006	60.8	15.5	80.0	131.6	5	92	8.0	90.0	35.0
cl011	67.4	41.1	74.3	205.9	5	22	16.6*	125.0	38.0
cl016	61.4	347.2	33.9	72.6	6	83	7.4	11.0–19.0	65.0–67.0
cl022	67.1	50.2	−3.2	40.9	3	490	5.6	305.0	71.0
cl022	63.4	271.0	31.9	9.1	3	49	17.8†	305.0	71.0
cl022	76.1	25.4	5.3	32.7	3	58	16.4‡	305.0	71.0
cl028	3.0	11.7	63.6	313.4	6	35	11.4†	128.0	84.0
cl028	62.2	29.4	33.4	222.8	5	26	15.3‡	128.0	84.0
cl034	39.5	298.4	45.6	39.0	5	39	12.4†	351.0	80.0
cl034	73.9	341.6	12.2	64.7	3	65	15.4‡	351.0	80.0
Sites without*†‡	66.9	19.1			4	44	14.0		
Sites without			61.0	68.0	4	2	82.9		
Wilbur Springs calcereous concentrations									
all sites comp. 1	63.3	8.2			6	75	8	see map	
all sites comp. 1			36.3	48.4	6	28	13		
all sites comp. 2	42.7	310.6			6	16	17	see map	
all sites comp. 2			55.7	359.0	6	61	9		
Stonyford volcanics									
sf034	78.3	9.5	68.6	34.6	6	51	9.5	330.0	12.0
sf040	78.9	337.7	71.1	8.2	6	67	8.3	310.0	11.0
sf046	86.8	159.4	34.5	63.8	6	121	6.1	330.0	56.0
sf052	83.4	308.2	36.2	52.4	5	30	14.2	330.0	56.0
sf057	77.3	349.2	53.7	31.1	7	32	10.9	320.0	28.5
sf065	72.8	344.3	52.9	24.2	5	50	11.0	321.0	27.0
sf071	79.2	355.3	39.5	51.8	6	45	10.1	335.0	46.0
All sites	80.8	347.2			7	133	5.3		
All sites			52.1	42.8	7	19	14.2		
Stonyford sediments									
sf010	80.8	332.2	42.2	68.1	3	249	7.8	350.0	50.0
sf015	81.1	32.5	47.1	56.4	4	77	10.5	333.0	35.0
sf020	82.1	322.5	47.8	66.3	5	78	8.7	347.0	45.0
sf025	74.5	15.3	25.2	56.0	4	65	11.5	340.0	55.0
sf029	80.4	354.2	32.1	59.0	4	83	10.2	340.0	55.0
sf100	61.3	332.2	42.0	30.7	6	104	6.6	340.5	45.0
sf106	66.1	333.8	49.3	32.4	5	106	7.5	340.5	37.0
sf112	65.6	310.5	50.5	33.5	3	91	13.0	338.5	46.0
sf118	67.2	320.1	52.1	33.0	5	221	5.2	339.5	40.0
sf124	52.5	329.9	40.8	50.2	4	19	21.9	3.5	75.0
sf128	22.7	308.9	59.4	26.9	5	122	7.0	7.5	75.0
sf134	35.3	305.4	51.7	57.8	4	35	15.7	7.5	75.0
sf138	69.8	327.1	38.7	50.1	4	549	3.9	345.0	55.0
sf142	74.4	292.0	49.0	62.7	5	372	4.0	346.5	53.0
all sites	67.0	323.3			14	17	9.8		
all sites			45.7	49.7	14	38	6.6		
Paskenta									
all samples	58.3	327.5			11	19	10.6	see map	
all samples			30.2	55.6	11	5	22.8		
Jacoby Butte									
jb001	67.2	355.9	56.6	99.6	4	279	5.5	40.0–56.0	46.0–57.0
jb020	68.3	29.2	7.9	327.3	7	45	9.1	215.5	79.0
jb027	79.2	22.9	0.8	320.3	4	84	10.1	220.0	86.0
jb031	71.7	10.8	40.4	268.3	4	26	18.3	158.0	58.0
jb035	56.6	4.7	79.0	137.4	4	227	6.1	83.0	42.0
jb039	55.0	348.5	63.9	99.1	3	47	18.1	49.0	52.0
all sites	67.0	5.3			6	54	9.2		
all sites			66.6	327.0	6	2	57.6		

In situ, field directions uncorrected for tilt. Corrected, field directions after tilt correction. Inc., inclination; Dec., declination; N, number of samples or sites averaged; κ, precision parameter; α_{95}, radius of 95% confidence in degrees. Comp. 1, low-temperature, low-coercivity component. Comp. 2, high-temperature, high-coercivity component. The direction of the dip is 90° clockwise from the direction of the strike.

*Site with high α_{95} excluded from mean of sites.

†High-coercivity components of samples with two components.

‡Low-coercivity components of samples with two components.

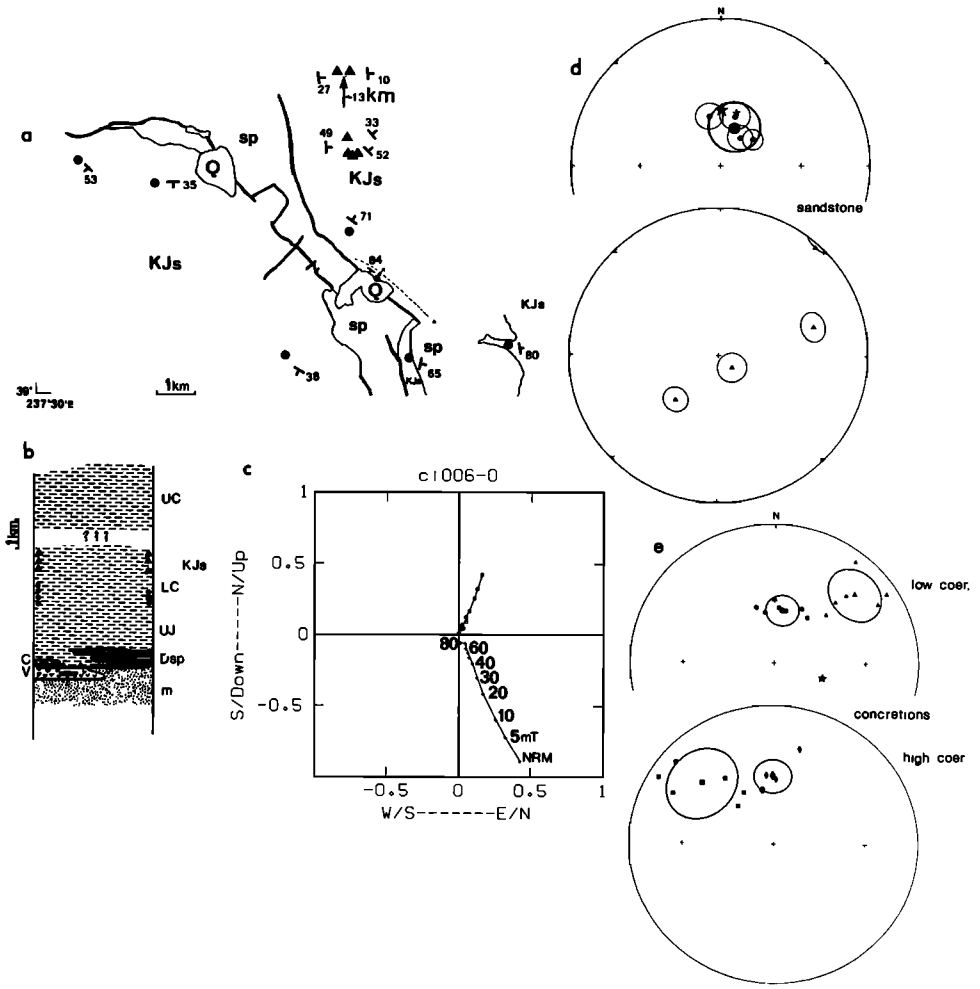


Fig. 3. Geologic setting and paleomagnetic results from Wilbur Springs area. (*a*) Generalized geologic map after *Rich* [1971]: Q, Quaternary; KJs, Great Valley sequence; sp, serpentinite. Thick lines are faults; solid circles are sandstone localities; solid triangles are calcareous concretion localities. Axis of antiform is marked by a dashed curve. (*b*) Part of stratigraphic column: m, melange; Dsp, Detrital serpentinite; V, volcanics; UJ, Upper Jurassic Great Valley sediments; LC, Lower Cretaceous Great Valley sediments; UC, Upper Cretaceous Great Valley sediments. Thick line with triangles, thrust fault. (*c*) See Figure 2*c*. Axial length 6.9×10^{-7} A m^2. (*d*) Equal area projection of paleomagnetic results from sandstones. Plus signs are at 45° inclination. Solid small circles, four site means before structural correction, with their ovals of 95% confidence; large full circle, the mean of the four site means, with its 95% confidence; solid triangles, site means after structural correction; large star, axial geocentric dipole field; small star, present-day magnetic field direction. Only four sites are plotted (other sites had bigger confidence ovals and were not included in the statistics). (*e*) Equal area projection of paleomagnetic results from calcareous concretions. Solid small circles and squares, site means before structural correction of low-coercivity and high-coercivity components, respectively; large full circle and square, the mean of the site means for both components; solid triangles and diamonds, site means of low- and high-coercivity components, respectively, after structural correction; star, axial geocentric dipole field on plot of low-coercivity component and expected 90 Ma field direction for North America [after *Irving and Irving*, 1982] on plot of high-coercivity component.

Wilbur Springs

The Wilbur Springs area was mapped by *Lawton* [1956] and more recently by *Rich* [1971], *Carlson* [1981], and *McLaughlin et al.* [1985] (Figure 3*a*). In this area, the Franciscan assemblage, the Coast Range thrust, and the lower part of the Great Valley sequence, including the basal ophiolite, are all folded around the southeast-plunging Wilbur Springs antiform. Upper Cretaceous rocks are not folded around this structure, but instead, appear to lie unconformably on the earlier deformed strata [*Rich*, 1971; *Carlson*, 1981] although, as was mentioned earlier, bedding-plane thrust faults cannot be ruled out. Sulfur springs, mercury deposits, and small intrusive bodies indicate Late Tertiary and Quaternary magmatic and related hydrothermal activity in this area.

Five or six samples (1–3 specimens each) were collected from each of seven sites in Lower Cretaceous sandstones of the Great Valley sequence around the entire antiform (Figures 3*a* and 3*b*). A pair of specimens from each site was cleaned by stepwise thermal and alternating magnetic field procedures. The results were very similar for the two procedures, therefore the rest of the samples were cleaned using alternating fields with peak fields ranging from 5 to 80 mT. Most samples had one component (Figure 3*c*), which on the basis of thermal demagnetization and isothermal remanent magnetization (IRM) experiments appears to reside in magnetite. The polarity of this component was normal in all specimens. Directions of magnetization were consistent within sites and between sites before correcting for the tilt of the beds and highly

scattered between sites after tectonic correction (Figure 3*d*, Table 1). The in situ direction is very close to the axial geocentric dipole field at the sampling locality; this remanent magnetization was clearly acquired after folding.

Samples in three of the sites had two magnetic components, one with high coercivity and high blocking temperature, the other with low coercivity and lower blocking temperature. IRM saturation experiments suggest that both hematite and magnetite are present. The high-coercivity components are quite consistent within each of the three sites but they are inconsistent between sites both before and after tectonic correction. The within-site scatter of these directions is higher than the within-site scatter for sites with one component. The low-coercivity component had an in situ direction similar to the in situ direction of the samples with only one magnetic component. The scatter was higher however, and therefore these data were not included in the mean. The magnetic direction residing in the magnetite may be a thermal or chemical overprint caused either by burial or by the recent volcanism in the Wilbur springs area. Volcanism in this area was as recent as 1.66 Ma and a hydrothermal hot water system is presently active in the area [*McLaughlin et al.*, 1985]. The alteration that produced the hematite might be related to the young volcanic and hydrothermal activity in this area. However, the magnetic direction of these components is not understood.

Calcareous concretions within the Upper Jurassic or Lower Cretaceous strata were also sampled near Wilbur Springs and $\approx$13 km north of Wilbur Springs. Thermal, alternating magnetic fields and the combination of thermal (to 400°C) and then alternating fields cleaning was used on these samples. The results of the three procedures were very similar. Most samples had two components; on the basis of the thermal and alternating field experiments, both components reside in magnetite. The polarities of both components were normal. The in situ direction of the lower-temperature (<400°C), lower-coercivity component is very close to the axial geocentric dipole field at the sampling site, and its scatter increases when a tectonic correction is applied. The higher-temperature, higher-coercivity component has an in situ magnetic direction of $D = 310.6$ and $I = 42.7$. The scatter of this component reduces significantly when tectonic correction is applied. The low-coercivity, low-temperature component could be the result of the young hydrothermal and volcanic activity, whereas the higher-temperature, higher-coercivity component might be the result of Late Cretaceous remagnetization followed by faulting within the sedimentary sequence.

Stonyford

Several fault-bounded areas near Stonyford contain relatively undeformed pillow lava, breccia, and interbedded radiolarian chert of Late Jurassic age [*Brown*, 1964; *Brown et al.*, 1981; *MacPherson*, 1983]. *Brown* [1964] mapped these volcanic rocks as lying between deformed Franciscan rocks and the overlying Jurassic shale and sandstone of the Great Valley sequence. *MacPherson* [1983], on the other hand, interprets the volcanic rocks as a seamount, that was subducted and metamorphosed as part of the Franciscan assemblage. We agree that the volcanic rocks probably formed as a seamount, but they are presently imbricated between the Franciscan and the Great Valley sequence and, based on petrographic similarities between the basalt and clasts in the basaltic sandstone in the lower part of the Great Valley sequence east of Stonyford, we would tend to agree with the interpretation of Brown, that at least the volcanic rocks along Stony Creek belong to the oceanic basement upon which the Great Valley sediments were deposited (Figure 4).

Seven sites were sampled in the pillow basalts along Stony Creek and 14 sites in the overlying basaltic sandstones (Figure 4). The attitudes of the volcanic rocks were measured using the interbedded radiolarian chert. Pairs of specimens were cleaned, one specimen thermally and the other by alternating magnetic fields, and a magnetic vector was fitted to the results using an algorithm modified after *Kirschvink* [1980]. The results from the magnetic and thermal experiments were similar in most samples. In some samples, however, the thermal cleaning exposed only one magnetic component, whereas the magnetic cleaning exposed two (Figure 4*c*). The lower-coercivity component has a direction similar to the component exposed by thermal cleaning. In order to recover both components, magnetic cleaning was used for the rest of the samples. Because the direction of the high-coercivity component was inconsistent within sites, both in the sedimentary and volcanic rocks, it was not included in the final analysis. The thermal demagnetization and IRM saturation experiments suggest that the lower-coercivity component resides in magnetite.

The polarity of the lower-coercivity component was normal and the direction was consistent within sites for both the volcanic and sedimentary rocks. The between-site consistency for the volcanic rocks was high before tectonic correction and low after correction (Figure 4*d*, Table 1). The in situ direction ($D = 347.2$, $I = 80.8$) is steeper than any expected direction for the North American craton for the Mesozoic and Cenozoic. For the sedimentary rocks the between site scatter was reduced when the site means were restored to paleohorizontal (Figure 4*e*, Table 1), suggesting that some of the deformation within the sedimentary rocks postdates the magnetization. However, the mean direction for the sedimentary rocks at the in situ orientation is $D = 323.0$ and $I = 67.0$, very close to the expected Late Cretaceous Noth American cratonal field at this location.

Paskenta

To the west and northwest of Paskenta, the Coast Range ophiolite widens to more than 5 km: on the west, it is emplaced against the Franciscan assemblage along a near-vertical fault; on the east, the Great Valley sequence locally rests depositionally on the ophiolite [*Bailey et al.*, 1970]. The Coast Range ophiolite and Great Valley sequence collectively form the Elder Creek terrane. The oldest fossils (Kimmeridgian) from the lowermost part of the Great Valley sequence in California were found here [*Jones*, 1975].

The Elder Creek terrane consists of, from west to east, a serpentinite-matrix melange, a dismembered sequence of ultramafic and gabbroic cumulate rocks, intrusive gabbro and diorite, a sheeted dike or sill complex, and basaltic lava, all unconformably overlain by ophiolitic breccia. Hornblende from the intrusive gabbro unit gave a K-Ar age of 166 $\pm$ 3 Ma [*McDowell et al.*, 1984], whereas hornblende from an intrusive diorite that cuts the cumulate ultramafic rocks gave a K-Ar age of 155 $\pm$ 5 [*Lanphere*, 1971 and oral communication, 1984]. These data plus recent geochemical analyses of the ophiolite [*Shervais and Kimbrough*, 1985; *Lagabrielle et al.*, 1986] suggest that the Coast Range ophiolite at Paskenta, as to the south near Del Puerto, consists of older oceanic (?) crust (compare 165 Ma) and a younger arc (compare 155 Ma) built on the older oceanic crust. All of these rocks are overlain,

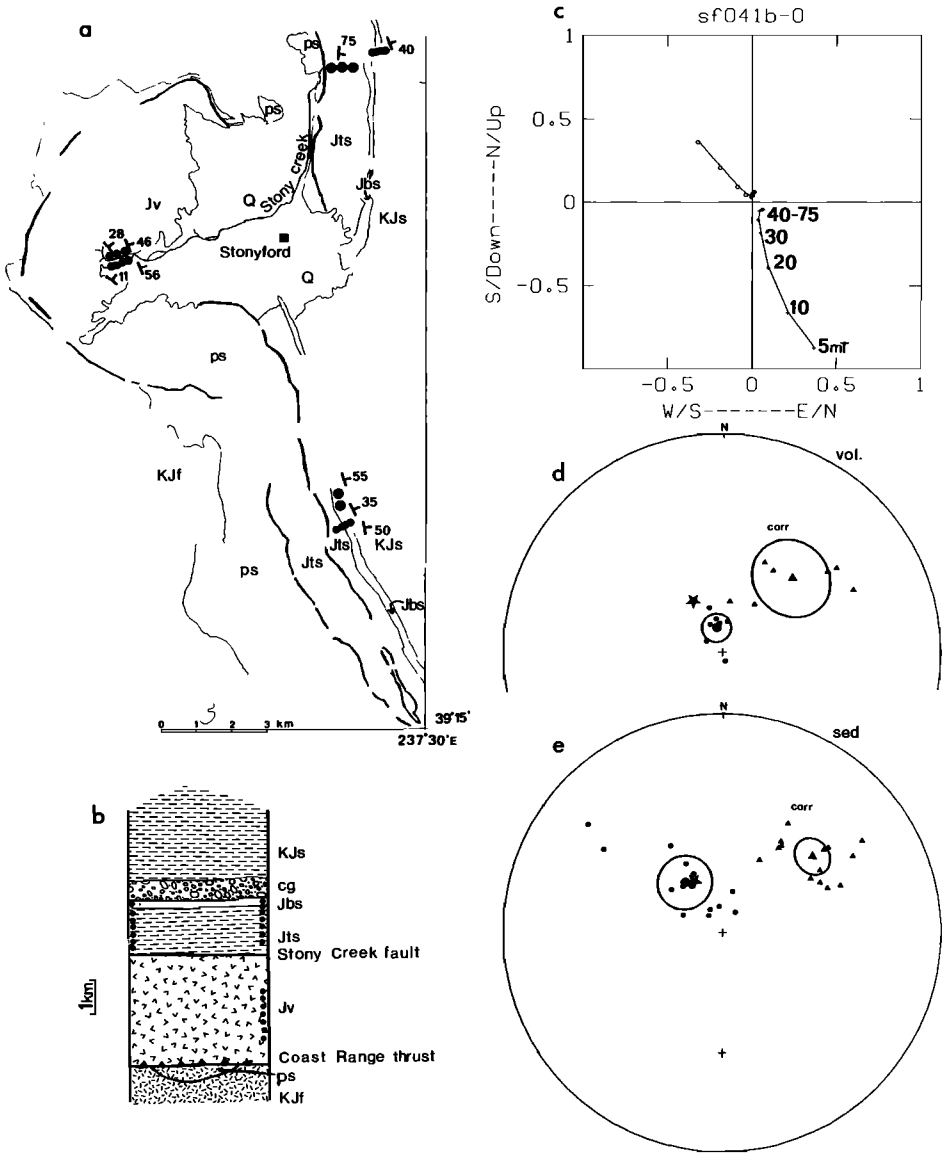


Fig. 4. Geologic setting and paleomagnetic results from the Stonyford area. (*a*) Generalized geology after *Brown* [1964]: KJs, Upper Jurassic and Lower Cretaceous Great Valley sequence; KJf, Jurassic and Cretaceous Franciscan rocks, Jbs, basaltic sandstone of the Great Valley sequence; Jts, Tuffaceous siltstone (Jurassic); Jv, Volcanic rocks; ps, Peridotite and serpentine; Q, Quaternary. (*b*) Part of the stratigraphic column sampled. Alphabetical symbols are the same as in Figure 4*a*; cg, Conglomerate. (*c*) Vector component plot of a sediment sample. Symbols are the same as in Figure 2*c*. Axial length 1.98×10^{-5} A m^2. (*d*) Equal area projection of the paleomagnetic results from volcanic rocks. Plus signs are at 45° and 90° inclination. Solid small circles, site means before structural correction; solid bigger circle, their mean with its 95% confidence; solid small triangles, site means after structural correction; solid large triangle, their mean with its 95% confidence oval; star, expected 90 Ma field direction after *Irving and Irving* [1982]. (*e*) Paleomagnetic results from the sedimentary rocks. Symbols are the same as in Figure 4*d*.

in turn, by the basal sedimentary rocks of the Great Valley sequence (Figures 5*a* and 5*b*).

The tectonic history recorded in the strata of the Elder Creek terrane is extremely complex (A. S. Jayko and M. C. Blake, Jr., unpublished manuscript, 1986). During the Late Jurassic ($\approx$154 Ma) the ophiolite was imbricated and uplifted prior to deposition of the ophiolitic breccia (the breccia lies depositionally across a thrust fault that emplaces cumulate ultramafic rocks onto dikes and volcanic rocks). Following deposition of the ophiolitic breccia, probably in the Late Cretaceous, the ophiolite and overlying breccia were cut by northwest-trending, left-lateral faults with a cumulative displacement of about 100 km. Steep, north-trending faults of probable Neogene age cut all of the earlier structures.

Samples collected as a pilot study in the Paskenta area were from different lithologies of the Jurassic Coast Range ophiolite. Most of the rocks strike about N-S and dip about 80° to the east. However, some internal deformation within the ophiolitic sequence enabled a test of the relative timing of deformation and magnetization. A few samples were taken from each of the following units: a volcanic clast several meters in diameter and calcareous concretions within the sandstones overlying the ophiolite; volcanic breccia; dikes (or sills); volcanic rocks from the serpentinite-matrix melange; intrusive gabbro; and relatively fresh cumulate ultramafic rocks. All samples were stepwise demagnetized thermally or in an alternating magnetic field (Figure 5*c*). The magnetic polarity of all samples was normal.

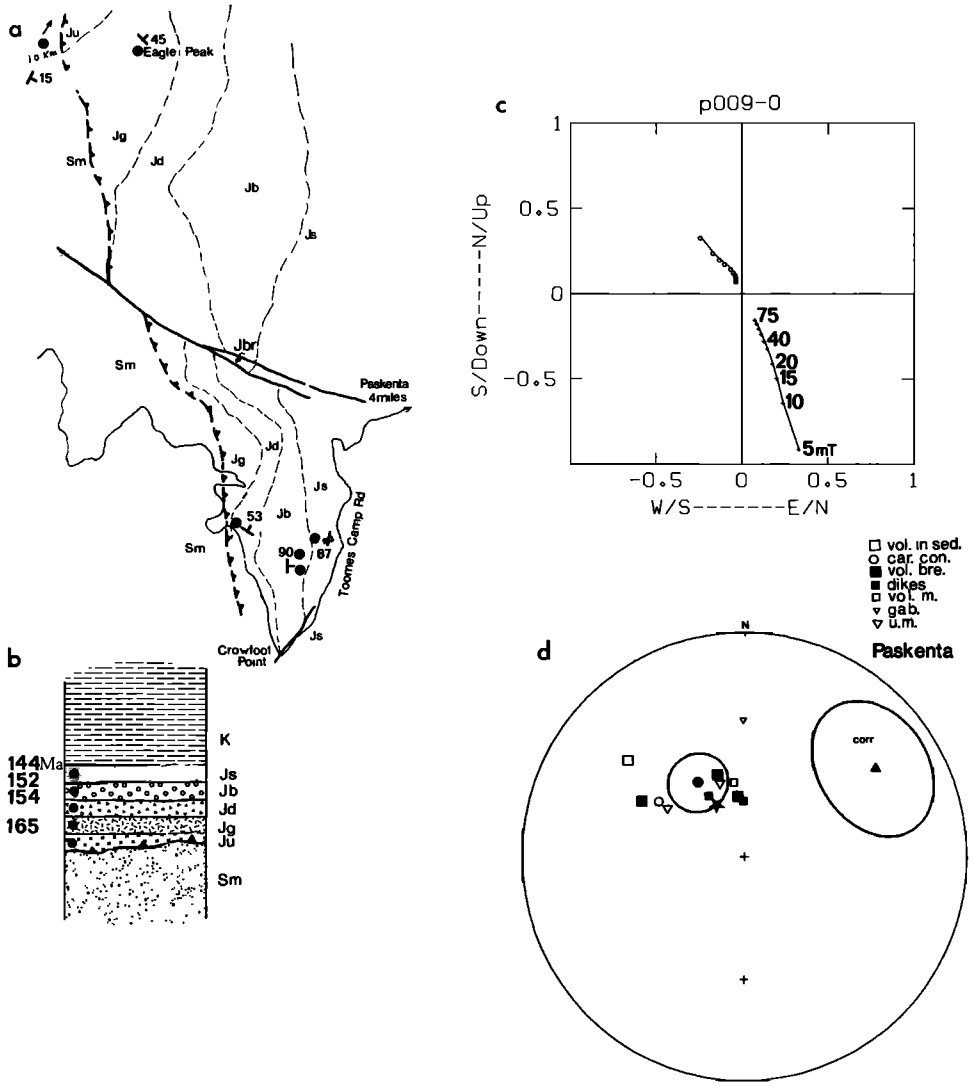


Fig. 5. Geologic setting and paleomagnetic results from the Paskenta area. (*a*) Generalized geology after M. C. Blake and A. S. Jayko (unpublished mapping, 1984, 1985, 1986): Js, Jurassic sedimentary rocks; Jb, Jurassic pillow lava and breccia; Jd, dike and sill complex; Jg, Jurassic intrusive gabbro; Ju, Jurassic ultramafic and gabbroic cumulates; Sm, Serpentinite melange; Thick lines are faults; thin lines are contacts; thick line with triangle shows thrust fault. (*b*) Part of the Stratigraphic column sampled. K, Cretaceous. (*c*) Vector component plot of a sample of a carbonate concretion. Symbols are the same as Figure 2*c*. Axial length 2.0×10^{-6} A m^2. (*d*) Equal area projection of the paleomagnetic results from Paskenta. Plus signs are at the 45° inclination. Field directions before structural correction of different rock units are as explained on figure; circle is their mean with its 95% confidence. Solid triangle, the mean after structural correction with its 95% confidence limit; star, expected 90 Ma field direction for North America [after *Irving and Irving*, 1982].

The mean in situ direction is very close to the Late Cretaceous expected direction for North America at this location (Figure 5*d*, Table 1). The scatter increases when the tectonic correction is applied, suggesting that the magnetization was acquired after the internal deformation of the ophiolitic sequence.

Jacoby Butte

A dismembered ophiolite sequence (with a Pb-U age of 169 ± 1 Ma; *Saleeby et al.* [1984]) overlain by lithic sedimentary rocks of the Upper Jurassic and Lower Cretaceous Riddle and Lower Cretaceous Days Creek Formations occurs at Jacoby Butte in southwest Oregon, [*Dott*, 1971; *Blake et al.*, 1985*b*] (Figures 6*a* and 6*b*). This sequence, which forms the Snow Camp terrane, has been correlated with the Coast Range ophiolite and overlying Great Valley sequence in California [*Jones*, 1973; *Blake et al.*, 1985*b*]. At Jacoby Butte the sedimentary strata are folded into a broad, north-trending syncline.

Samples were collected from six sites on both limbs of the fold in the Lower Cretaceous Days Creek Formation through the time-stratigraphic interval of the Valanginian and Hauterivian (≈138–125 Ma). Stepwise thermal and magnetic cleaning of one or two pairs of specimens from each site gave very similar results. Alternating magnetic fields were used for the rest of the specimens. Some samples had two magnetic components: a low-temperature, low-coercivity normal component and a high-temperature, high-coercivity reversed component (Figure 6*c*). IRM saturation experiments exclude the existence of hematite, Since the reverse direction was of very low inten-

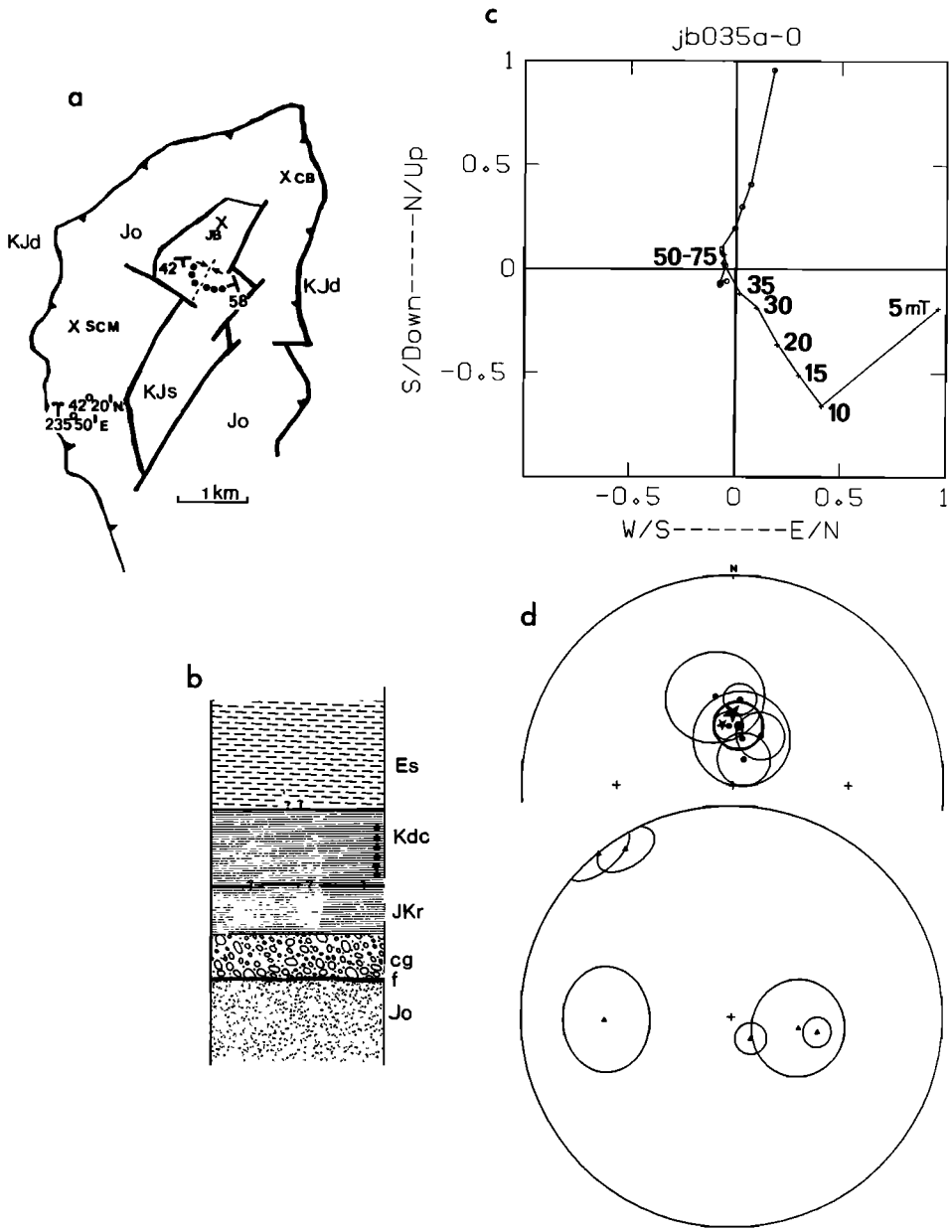


Fig. 6. Geologic setting and paleomagnetic results from the Jacoby Butte area, southwest Oregon. (*a*) Generalized geology after *Blake et al.* [1985*a*]; KJd, Dothan Formation; Jo, ophiolite; KJs, Riddle and Days Creek Formations; JB, Jacoby Butte; CB, Collier Butte; SCM, Snow Camp Mountain; dashed line, fold axis. (*b*) Part of the stratigraphic column sampled: cg, conglomerate; KJr, Riddle Formation; Kdc, sedimentary rocks of Days Creek Formation; Es, Eocene overlap asemblage; f, fault. (*c*) See Figure 2*c*. Axial length 1.7×10^{-8} A m^2. (*d*) Equal area projection of the paleomagnetic results. Plus signs are at 45° inclination. Solid circles and triangles are the same as for Figure 3*d*. Large star, axial geocentric dipole; small star, 50 Ma field direction for North America as calculated from *Irving and Irving* [1982].

sity and was present in only a few samples, its mean orientation could not be well determined. The in situ site means all plot near the axial geocentric dipole field at this locality. The scatter increases considerably for the site mean directions when they are restored to paleohorizontal (Figure 6*d*, Table 1), suggesting magnetization after folding.

Discussion

At the five localities sampled, the paleomagnetic results suggest complete magnetic overprinting; these results are summarized in Figure 7. The following discussion bears on the age and source of the measured magnetization and the implications of the paleomagnetic results for the tectonics of northern California and southwest Oregon.

At the Del Puerto, Wilbur Springs, and Jacoby Butte localities the axial dipole direction of the magnetic remanence suggests Late Tertiary or Quaternary remagnetization. At Del Puerto and Wilbur Springs only normal polarity was found, whereas at Jacoby Butte both normal and reversed components are present. The in situ paleomagnetic directions found in the Stonyford and Paskenta areas and in the calcareous concretions near Wilbur Springs are not parallel to the Late

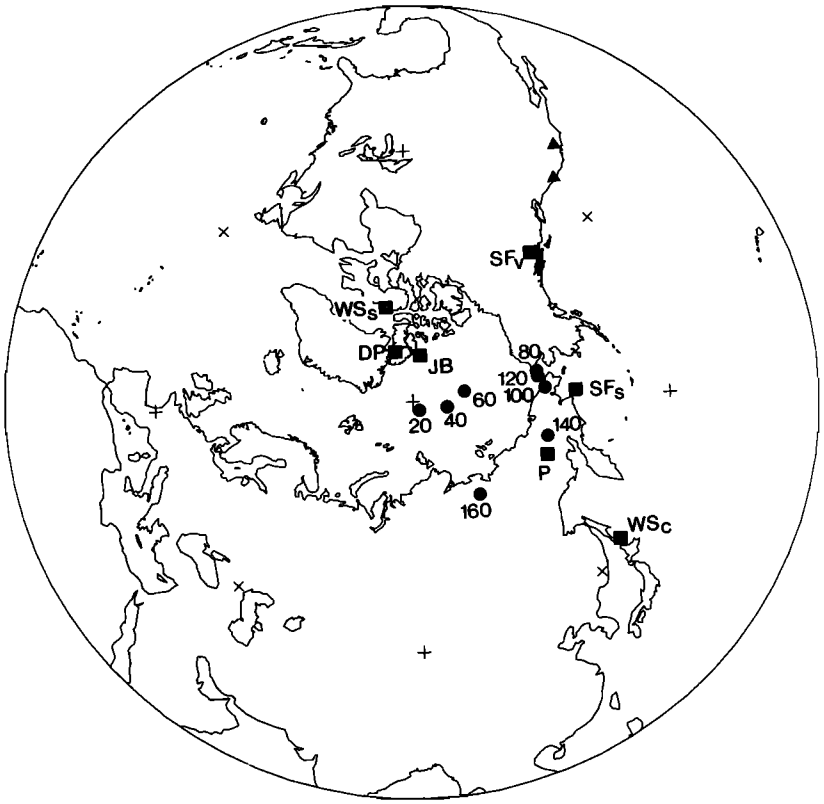


Fig. 7. Summary of paleomagnetic results. Two triangles are the northern and southern sampling localities. Solid small circles are the pole positions for cratonal North America since 160 Ma, every 20 m.y. [after *Irving and Irving*, 1982]. Numerals are in million years. Squares are the mean pole positions as calculated from the in situ directions at the five sites studied. DP, Del Puerto; JB, Jacoby Butte; P, Paskenta; SFv, pole from the volcanic rocks at Stonyford; SFs, pole from the sedimentary rocks at Stonyford; WSc, Wilbur Springs concretions (high-coercivity components); WSs, Wilbur Springs sandstones.

Tertiary magnetic field. At these three localities only normal polarities were found. The magnetization at these localities may be of Late Cretaceous age, in agreement with other results from the northern Great Valley [*Mankinen*, 1978; *Mankinen and Irwin*, 1982].

Origins of Remagnetization

The Upper Jurassic to Upper Cretaceous Great Valley sequence has an apparent exposed thickness of more than 15 km along the west side of the Sacramento Valley [*Dickinson et al.*, 1969]. Similarly, in the Del Puerto area the measured thickness of the Upper Cretaceous sequence is close to 10 km [*Maddock*, 1964]. However, geophysical data [*Cady*, 1975; *Wentworth et al.*, 1984] suggest maximum depths to basement of only 6–8 km. The present temperature gradients in the Great Valley range between 2.2°C/100 m and 3.6°C/100 m [*Zieglar and Spotts*, 1978], suggesting temperatures of 150°–270°C at the base of the sequence on the west side of the Valley. *Dickinson et al.* [1969] describe the mineral assemblages at Cache Creek on the northwest side of the valley and suggest burial metamorphism as the origin for laumontite in the Lower Cretaceous rocks. Formation of laumontite suggests temperatures of up to 185°C [*Cho et al.*, 1984]. *Bailey and Jones* [1973] studied localities in the northwestern Great Valley where the alteration is to prehnite-pumpellyite facies, suggesting temperatures of ≈200°C. Both studies suggest a thermal gradient of 2°–3°C/100 m. *Suckecki and Land* [1983] studied oxygen isotopic data from anthigenic phases in the northwestern Great Valley and concluded the temperature gradient was about 25°C/km during maximum burial, which was to about 6–7 km. These studies suggest that during the Cretaceous the temperature at the base of the sequence on the west side of the valley was about 150°–200°C for several tens of millions of years, from the time most of the sequence was deposited until the sedimentary rocks were uplifted and tilted in the Late Cretaceous or Tertiary.

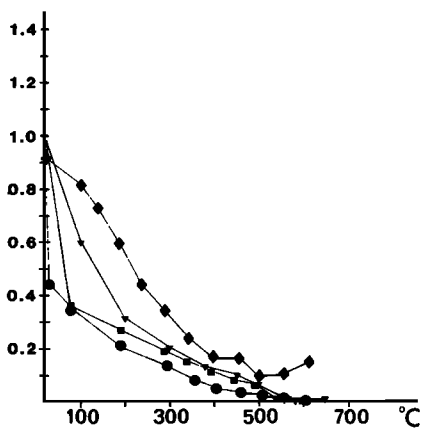


Fig. 8. Normalized intensity during thermal demagnetization experiments. Triangles, sample sf018-1 from sediments at Stonyford; squares: sample sf044-1 from volcanics at Stonyford, circles, sample cl011-1 from sandstone at Wilbur Springs; diamonds, sample ws002-0 from calcareous concretions from Wilbur Springs.

The rocks studied have a very wide range of unblocking temperatures. Examples of the behavior of the magnetic intensity in a thermal demagnetizing experiment are given in Figure 8. The theoretical curves for magnetite of *Pullaiah et al.* [1975, Figure 1*a*] suggest that an unblocking temperature of about 350°C in a laboratory experiment of about 5 min length is equivalent to heating to about 180°C for 10 m.y. However, both *Walton* [1980] and *Dunlop* [1983] conclude that these curves underestimate the reheating temperature needed in short-term thermal demagnetizations experiments of a magnetization acquired during long-term heating at lower temperatures. These conclusions together with the wide range of unblocking temperatures of the rocks studied suggest that heating of the Jurassic rocks to about 200°C for 10 million years could result in unblocking of the magnetization. In such a case a thermoremanent magnetization would be blocked in these units during cooling caused by uplift. The timing at which the magnetization is acquired can vary as a result of varying thickness, time of uplift, and local thermal gradient.

At temperatures of ≈200°C, magnetite could be formed as a result of decomposition of hornblende (J. G. Liou, personal communication, 1984). In this case, the magnetization could be in magnetite formed chemically after deposition, but the magnetization would be blocked only during uplift and cooling. It is not known if such reactions took place in the rocks studied. Mineralogical and petrological studies could determine whether the magnetite is depositional in origin or whether it resulted from chemical reactions. Whatever the source of the magnetite in these rocks, it seems likely that the magnetization in the basal northwestern part of the Great Valley sequence was acquired during their uplift and cooling.

S. Graham (personal communication, 1985) suggests, however, that based on vitrinite reflectance data, the temperatures at or near the base of the Great Valley sequence were, in general, less than 100°C and that the rocks could have been heated to higher temperatures only for a very short time period. If this were the case, the mineralogical and isotopic data discussed above has to be reevaluated in light of temperatures of 150°–200°C affecting the rocks only for a very short time, and chemical remagnetization residing in magnetite and not heat would be the source of observed magnetic components (chemical remagnetization in carbonates was described by *McCabe et al.* [1983]).

Hydrothermal activity associated with volcanism and tectonic events can be a source of localized chemical remagnetization. In such cases, the remagnetization could postdate remagnetization as a result of burial and uplift.

The good agreement of the different radiometric ages (U-Pb, Ar-Ar, and K-Ar on amphiboles) and the fossil data available from the ophiolitic and overlying rocks suggest that the chemical or thermal processes that affected the magnetization did not reset the isotopic ages. The exact temperatures at which the different isotopic systems are closed and reopened, and the influence of the length of time during which the rocks are heated is not yet well known [*Cliff*, 1985]. Therefore the constraints implied by this data on the length of time and temperatures or the chemical reactions that took place can not be evaluated at present.

Interpretation of Paleomagnetic Results

The results from Del Puerto suggest that magnetization was acquired after folding. The in situ direction is very near the expected magnetic direction for the late Cenozoic, which is consistent with magnetization during the Miocene to Holocene uplift of the Diablo Range, presumably in response to activity along the San Andreas system [*Page*, 1981; *Aydin and Page*, 1984].

The results from the sandstones and the lower-coercivity component in the calcareous concretions in the Wilbur Springs area similarly suggest magnetization after folding. The in situ directions of magnetization are near the expected late Cenozoic direction. The measured magnetization could be due to the Pliocene and Pleistocene volcanic and hydrothermal activity in the area.

The normal magnetic vector at Jacoby Butte failed the fold test and the in situ direction is consistent at the 95% confidence limit with the Tertiary axial geocentric dipole. Since the total thickness of the sedimentary rocks here is only a few hundred meters, burial followed by uplift can be excluded as the cause of remagnetization. The remagnetization, probably chemical, may have occurred during a major Early Tertiary tectonic event in this area, the accretion of the Siletz terrane [*Simpson and Cox*, 1977; *Magill and Cox*, 1980].

Interpretation of the paleomagnetic results from Stonyford and Paskenta and the higher-coercivity component from the calcareous concretions from the Wilbur Springs area depends on the relative timing of magnetization and tilting. All of the paleomagnetic directions from these three localities have normal polarities. The simplest interpretation based only on the paleomagnetic results from Paskenta (Figure 5*d*) is remagnetization during the long normal polarity in the Late Cretaceous, postdating deformation. This would imply that all the eastward tilting (≈80°) predates the remagnetization, a fact not supported by the geologic relations described above. The simplest interpretation of the sequence of events, suggested by the paleomagnetic results from Stonyford, is similar. The volcanic rocks were remagnetized after deformation (Figure 4*d*). The in situ direction is too steep for a Late Cretaceous field direction and could imply some postremagnetization tilting of the whole block. The only normal polarity found in the in situ paleomagnetic results from the Stonyford sedimentary rocks (Figure 4*e*) are in agreement with remagnetization during the Late Cretaceous. However, the scatter reduces when a tectonic correction is applied, suggesting some postremagnetization deformation. The results from the high-coercivity component from the calcareous concretions from the Wilbur Springs area similarly suggest remagnetization of these rocks during a normal polarity interval, followed by postremagnetization deformation.

This simple interpretation of the sequence of events at Paskenta can be summarized as follows: (1) formation of the ophiolite and initial magnetization, with subsequent faulting (this faulting was followed by deposition of ophiolitic breccia and the basal sedimentary rocks, all in the Late Jurassic and, (2) tilting of the ophiolite and overlying sedimentary rocks, followed by remagnetization in the Late Cretaceous. At Stonyford this interpretation would imply the following sequence of events: (1) formation of pillow basalts and initial magnetization with subsequent faulting (this faulting was followed by deposition of the basal part of the Great Valley sequence, all in the Late Jurassic); (2) tilting of the ophiolite and overlying sedimentary rocks, followed by remagnetization in the Late Cretaceous; and (3) Pliocene tilting of the pillows as a unit and some disruption within the sedimentary rocks at Stonyford.

An alternative interpretation of the paleomagnetic results of Stonyford, Paskenta and the high-coercivity component from the calcareous concretions in Wilbur Springs area is that the

eastward tilting is all younger than the magnetization. In this case, some deformation within the ophiolite (at Paskenta) and within the volcanics at Stonyford predating remagnetization is implied. In addition, some clockwise rotation prior to or during tilting would be implied, for any time of remagnetization, Cretaceous through Tertiary. This possibility is consistent with widespread clockwise rotations along the northwest coast of the United States [*Beck*, 1980]. A Late Cretaceous age of the remagnetization is preferred because only normal magnetizations were found.

The more complicated sequence of events, including eastward tilting both before and after remagnetization, is in better agreement with the known geology. However, the implications of this model depend upon the distribution of tilting through time which is poorly constrained. Other interpretations of the sequence of events in the northern California Coast Ranges can not be excluded; however, they call for a different history at each of the three locations studied.

SUMMARY

Paleomagnetic results from sedimentary rocks from the lower part of the Great Valley sequence overlying the Coast Range ophiolite at Del Puerto, Wilbur Springs, and Jacoby Butte suggest a Cenozoic remagnetization. At Del Puerto this remagnetization could be the result of deep burial for a long period followed by cooling as a result of uplift or chemical remagnetization in the Tertiary. At Wilbur Springs, a possible cause of remagnetization is Late Tertiary or Quaternary chemical alteration related to hydrothermal activity.

The remagnetization at Paskenta, Stonyford and the high-coercivity component from the calcareous concretions at Wilbur Springs could have occurred from the Late Cretaceous through Tertiary. Unblocking of the original magnetization could have been caused by burial to 7 km or more for tens of millions of years. Cooling and blocking of the magnetization would then be a result of uplift, which in this area could have taken place either in the late Early to Late Cretaceous with minor faulting in the Late Cenozoic, or more likely starting in the Cretaceous and continuing into the Tertiary. In the latter case eastward tilting was accompanied by clockwise rotation. However, chemical remagnetization before, during, or after tilting can not be excluded.

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REFERENCES

Aydin, A., and B. M. Page, Diverse Pliocene-Quaternary tectonics in a transform environment, San Francisco Bay region, California, *Geol. Soc. Am. Bull.*, *95*, 1303–1317, 1984.

Bailey, E. H., and D. L. Jones, Metamorphic facies indicated by vein minerals in basal beds of the Great Valley sequence, northern California, *J. Res. U.S. Geol. Surv.*, *1*, 383–385, 1973.

Bailey, E. H., M. C. Blake, Jr., and D. L. Jones, On-land Mesozoic oceanic crust in California Coast Ranges, *U.S. Geol. Surv. Prof. Pap.*, *700-C*, C70–C81, 1970.

Beck, M. E., Jr., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. Geophys. Res.*, *85*, 7115–7131, 1980.

Beebe, W., and B. P. Luyendyk, Preliminary paleomagnetic results from the Llanda remnant of the Coast Range Ophiolite, San Benito County, California, *Eos Trans. AGU*, *65*(45), 686, 1983.

Berkland, J. O., Rice Valley outlier—New sequence of Cretaceous—Paleocene strata in northern California Coast Ranges, *Geol. Soc. Am. Bull.*, *84*, 2389–2406, 1973.

Blake, M. C., Jr., Geologic transect of the northern Diablo Range, California, in *Upper Mesozoic Franciscan and Great Valley Sequence, Central Coast Range, California*, edited by V. Frizzell, pp. 35–43, S.E.P.M., Los Angeles, Calif., 1981.

Blake, M. C., Jr., and D. L. Jones, The Franciscan assemblage and related rocks in northern California: A reinterpretation, in *The Geotectonic Development of California, Rubey vol. 1*, edited by W. G. Ernst, pp. 307–328, Prentice-Hall, Englewood Cliffs, N.J., 1981.

Blake, M. C., Jr., D. G. Howell, and D. L. Jones, Preliminary tectonostratigraphic terrane map of California, *U.S. Geol. Surv. Open File Rep.*, *82-593*, 1982.

Blake, M. C., Jr., D. S. Harwood, E. Helley, W. P. Irwin, A. S. Jayko, and D. L. Jones, Preliminary geologic map of the Red Bluff 1:100,000 quadrangle, *U.S. Geol. Surv. Open File Rep. 84-105*, 1984.

Blake, M. C., Jr., D. C. Engebretson, A. S. Jayko, and D. L. Jones, Tectonostratigraphic terranes, in southwest Oregon, in *Tectonostratigraphic Terranes of the Circum Pacific Region, Earth Sci. Series*, vol. 1, edited by D. G. Howell, pp. 147–157, Circum Pacific Council for Energy and Mineral Resource, Houston, Tex., 1985*a*.

Blake, M. C., Jr., A. S. Jayko, and R. J. McLaughlin, Tectonostratigraphic terranes of the northern Coast Ranges, California, in *Tectonostratigraphic Terranes of the Circum Pacific Region, Earth Sci. Series*, vol. 1, edited by D. G. Howell, pp. 159–171, Circum Pacific Council for Energy and Mineral Resource, Houston, Tex., 1985*b*.

Brown, R. D., Jr., Geologic map of the Stonyford quadrangle, Glenn, Colusa and Lake Counties, California, *U.S. Geol. Surv. Miner. Invest. Field Stud., MF-279*, 1964.

Brown, R. D., Jr., D. J. Grimes, R. Leinz, F. E. Federspiel, and A. M. Leszcykowsky, Mineral resources of the Snow Mountain wilderness study area, California, *U.S. Geol. Surv. Bull.*, *1495*, 48 pp., 1981.

Cady, J. W., Magnetic and gravity anomalies in the Great Valley and western Sierra Nevada metamorphic belt, California, *Spec. Pap. Geol. Soc. Am.*, *168*, 56 pp., 1975.

Carlson, C., Sedimentary serpentinites of the Wilbur Springs area—A possible early Cretaceous structural and stratigraphic link between the Franciscan complex and the Great Valley sequence, Masters thesis, 105 pp., Stanford Univ., Stanford, Calif., 1981.

Cho, M., J. G. Liou, and S. Maruyama, Transition from the zeolite to prehnite-pumpellyite facies in the Karmutsen metabasites, Vancouver Island, *Geol. Soc. Am. Abstr. Programs*, *16*(6), 470, 1984.

Cliff, R. A., Isotopic dating in metamorphic belts, *J. Geol. Soc. London*, *142*, 97–110, 1985.

Dickinson, W. R., R. W. Ojakangas, and R. J. Stewart, Burial metamorphism of the late Mesozoic Great Valley sequence, Cache Creek, California, *Geol. Soc. Am. Bull.*, *80*, 519–526, 1969.

Dickinson, W. R., R. V. Ingersoll, and S. A. Graham, Paleogene sediment dispersal and paleotectonics in northern California, *Geol. Soc. Am. Bull.*, *90*, 1458–1528, 1979.

Dott, R. H., Jr., Geology of the southwestern Oregon coast west of the 124th meridian., *Bull. Oreg. Dep. Geol. Mineral Ind.*, *69*, 63 pp., 1971.

Dunlop, D. J., Viscous magnetization of 0.04–100 μm magnetites, *Geophys. J. R. Astron. Soc.*, *74*, 667–687, 1983.

Evarts, R. C., The geology and petrology of the Del Puerto ophiolite, Diablo Range, Central California Coast Range, North American Ophiolites, pp. 121–140, *Bull. 95*, State of Oreg. Dep. of Geol. and Miner. Ind., Portland, Oreg., 1977.

Griscom, A., The contact between the Great Valley sequence and the Franciscan assemblage California, from magnetic data (abstract), *Eos Trans. AGU*, *64*(45), 867–868, 1983.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and A. R. Walters, *A Geologic Time Scale*, 128 pp., Cambridge University Press, New York, 1982.

Hopson, C. A., J. M. Mattinson, and E. A. Pessagno, Jr., Coast Range Ophiolite, western California, in *The Geotectonic Development of California, Rubey vol. 1*, edited by W. G. Ernst, pp. 419–510, Prentice-Hall, Englewood Cliffs, N. J., 1981.

Ingersoll, R. V., Initiation and evolution of the Great Valley forearc basin of northern and central California, U.S.A., Trench Forearc Geology: Sedimentation and Tectonics on Modern and Ancient Active Plate Margins, pp. 459–467, *Spec. Publ. 10*, Geological Society of London, 1982.

Irving, E., and G. A. Irving, Apparent Polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, *Geophys. Surv.*, *5*, 141–188, 1982.

Jones, D. L., Structural significance of upper Mesozoic biostratigraphic units in northern California and southwestern Oregon, *Geol. Soc. Am. Abstr. Programs*, *5*(7), 684–685, 1973.

Jones, D. L., Discovery of Buchia rugosa of Kimmeridgian age from the base of the Great Valley sequence (abstract), *Geol. Soc. Am. Abstr. Programs*, *7*(3), 330, 1975.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.

Lagabrielle, Y., F. Roure, A. Coutelle, R. Maury, J. L. Joron, and P. Thonon, The Coast Range ophiolites (northern California): Possible arc and basin remnants back arc; Their relations with the Nevadan Orogeny, *Bull. Geol. Soc. France*, *8*(6), 981–999, 1986.

Lanphere, M. A., Age of the Mesozoic oceanic crust in the California Coast Ranges, *Geol. Soc. Am. Bull.*, *82*, 3209–3212, 1971.

Lawton, J. E., Geology of the north half of the Morgan Valley quadrangle and the south half of the Wilbur Springs quadrange, Ph.D. dissertation, Stanford Univ., Stanford, Calif., 1956.

Luyendyk, B. P., and J. S. Hornafius, Paleolatitude of the Point Sal Ophiolite, *Geol. Soc. Am. Abstr. Programs*, *14*(4), 182, 1982.

MacPherson, G. J., The Snow Mountain volcanic complex: an onland seamount in the Franciscan terrain, California, *J. Geol.*, *91*, 73–92, 1983.

Maddock, M. E., Geology of the Mt. Boardman quadrangle, Santa Clara and Stanislaus counties, California, Map Sheet 3, scale 1:62, Calif. Div. Mines and Geology, Sacramento, Calif., 1964.

Magill, J. R., and A. Cox, Tectonic rotation of the Oregon western Cascades, *Spec. Pap. 10*, 67 pp., Dep. of Geol. and Miner. Ind., State of Oreg., Portland, 1980.

Mankinen, E. A., Paleomagnetic evidence for a late Cretaceous deformation of the Great Valley sequence, Sacramento valley, California, *J. Res. U.S. Geol. Surv.*, *6*(3), 383–390, 1978.

Mankinen, E. A., and W. P. Irwin, Paleomagnetic study of some Cretacous and Tertiary sedimentary rocks of the Klamath Mountains province, California, *Geology*, *10*, 82–87, 1982.

McCabe, C., R. Van der Voo, D. R. Peacor, C. R. Scotese, and R. Freeman, Diagenetic magnetite carries ancient yet secondary remanence in some Paleozoic sedimentary carbonates, *Geology*, *11*, 221–223, 1983.

McDowell, F. W., D. H. Lehman, P. R. Gucwa, D. Fritz, and J. C. Maxwell, Glaucophane schists and ophiolites of the northern California Coast Ranges: Ages, mineralogy, and their tectonic implications, *Geol. Soc. Am. Bull.*, *95*, 1373–1382, 1984.

McLaughlin, R. J., and H. N. Ohlin, Tectonostratigraphic framework of the Geysers—Clear Lake region, California, in *Franciscan Geology of Northern California*, edited by M. C. Blank, Jr., pp. 135–148, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1984.

McLaughlin, R. J., H. N. Ohlin, D. J. Thormahlen, D. L. Jones, J. W. Miller, and C. D. Blome, Geologic Map and structure sections of the Little Indian Valley—Wilbur Springs geothermal area, northern Coast Ranges, California, *U.S. Geol. Surv. Open File Rep.*, *85-285*, 1985.

McWilliams, M. O., and D. G. Howell, Exotic terranes of western California, *Nature*, *297*, 215–217, 1982.

Page, B. M., The southern Coast Ranges, in *The Geotectonic Development of California, Rubey vol. 1*, edited by W. G. Ernst, pp. 329–417, Prentice-Hall, Englewood Cliffs, N. J., 1981.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett.*, *28*, 133–143, 1975.

Raymond, L. E., Franciscan geology of the Mount Oso area, California, Ph.D. thesis, 180 pp., Univ. of Calif., Davis, 1973.

Ricci, M. P., E. M. Moores, K. L. Verosub, and J. S. McClain, Geologic and gravity evidence for thrust emplacement of the Smartville ophiolite, *Tectonics*, *4*, 539–546, 1985.

Rich, E. I., Geologic map of Wilbur Springs quadrangle Colusa and Lake counties, California, 1:48,000, *U.S. Geol. Surv. Misc. Geol. Invest. Map, I-538*, 1971.

Saad, A. H., Paleomagnetism of Franciscan ultramafic rocks from Red Mountain, California, *J. Geophys. Res.*, *74*, 6567–6578, 1969.

Saleeby, J., Ocean floor accretion and volcanoplutonic arc evolution of the Mesozoic Sierra Nevada, in *The Geotectonic Development of California, Rubey vol. 1*, edited by W. G. Ernst, pp. 132–181, Prentice-Hall, Englewood Cliffs, N. J., 1981.

Saleeby, J. B., M. C. Blake, Jr., and R. G. Coleman, Pb/U zircon ages on thrust plates of the west central Klamath Mountains and Coast Ranges, northern California and southern Oregon, *Eos Trans. AGU*, *65*(45), 1147, 1984.

Schweickert, R. A., and D. S. Cowan, Early Mesozoic tectonic evolution of the western Sierra Nevada, California, *Geol. Soc. Am. Bull.*, *86*, 1329–1336, 1975.

Sharp, W. D., and R. C. Evarts, New constraints on the environment of formation of the Coast Range Ophiolite at Del Puerto Canyon, California, *Geol. Soc. Am. Abstr. Programs*, *14*(4), 233, 1982.

Shervais, J. W., and D. L. Kimbrough, Geochemical evidence for the tectonic setting of the Coast Range ophiolite: A composite island-arc-oceanic crust terrane in western California, *Geology*, *13*, 35–38, 1985.

Simpson, R. W., and A. Cox, Paleomagnetic evidence for tectonic rotation of the Oregon Coast Range, *Geology*, *5*, 585–589, 1977.

Suckecki, R. K., and L. S. Land, Isotopic gechemistry of burial-metamorphosed volcanogenic sediments, Great Valley sequence, northern California, *Geochim. Cosmochim. Acta*, *47*, 1487–1499, 1983.

Suppe, J., Geological map and cross section of the southern part of the northern Coast Ranges and Sacramento Valley, California, *Map Chart Ser. Geol. Soc. Am., MC-28B*, 1978.

Suppe, J., Structural interpretation of the southern part of the northern Coast Ranges and Sacramento Valley, California: Summary, *Geol. Soc. Am. Bull.*, *90*, 327–330, 1979.

Walton, D., Time-temperature relations in the magnetization of assemblies of single domain grains, *Nature*, *286*, 245–247, 1980.

Ward, P. D., K. L. Verosub, and J. W. Haggart, Marine magnetic anomaly 33–34 identified in the Upper Cretaceous of the Great Valley sequence of California, *Geology*, *11*, 90–93, 1983.

Wentworth, C. M., M. C. Blake, Jr., D. L. Jones, A. W. Walter, and M. D. Zoback, Tectonic wedging associated with emplacement of the Franciscan assemblage, California Coast Ranges, in *Franciscan Geology of Northern California*, edited by M. C. Blake, Jr., pp. 163–173, Pacific Section, Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1984.

Williams, K. M., The Mount Diablo ophiolite, Contra Costa county, California, Masters thesis, San Jose State Univ., San Jose, Calif., 1983.

Zieglar, D. L., and J. H. Spotts, Reservoir and source-bed history and Great Valley, California, *Am. Assoc. Pet. Geol. Bull.*, *62*, 813–826, 1978.

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